Investigating Grokking in Modular Arithmetic

Arianna Paone, Damien Bardina

June 2, 2026

1 Introduction

Grokking, introduced by Power et al. [2], describes a training phenomenon in which a neural network first memorises the training set and only after prolonged optimisation suddenly achieves near-perfect generalisation on held-out data. This delayed transition suggests that networks may internally switch between qualitatively different solution strategies during training.

Modular arithmetic provides a clean setting for studying grokking: the task is simple, the full dataset can be exhaustively generated, and correct generalisation requires learning mathematical structure rather than memorising individual examples. In this project we train a transformer architecture on $(a + b + c) \bmod 37$, study how weight decay and training fraction affect grokking dynamics, and perform a mechanistic analysis of the learned algorithm, inspired by the interpretability methodology of Nanda et al. [1].

2 Background

Grokking and regularisation:

Power et al. [2] showed that grokking is strongly driven by weight decay. Without sufficient regularisation, models rely on memorisation strategies that do not generalise, if the algorithmic solution is not immediate. With strong weight decay the optimiser is biased toward low-norm solutions, eventually discovering a compact algorithmic representation. Smaller training fractions increase the delay before generalisation without changing final accuracy, because the model memorises the data quickly but takes longer to compress its solution under regularisation pressure.

Fourier mechanisms:

Nanda et al. [1] showed that one-layer transformers trained on $(a + b) \bmod p$ do not learn to solve modular addition by memorising input-output pairs. The key insight is that modular arithmetic is periodic: the integers $0, 1, \ldots, p-1$ are not a line but a circle (in modular arithmetics), and any function on them can be decomposed into discrete Fourier modes. A transformer that represents each number t as the pair $(\cos(2\pi kt/p), \sin(2\pi kt/p))$ for some frequency k can then exploit the angle addition identity

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta, \qquad \sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

to compute the Fourier representation of $a+b$ from the Fourier representations of a and b individually, with the periodicity of cosine and sine handling the modular wrap without any explicit remainder operation. This computation requires multiplication of Fourier components, which is why a nonlinear network is necessary.

The reason a transformer moves toward this solution under weight decay is that it is simpler: rather than storing a separate rule for each of the p^2 input pairs, the model needs only to learn the Fourier embedding of p numbers and a smaller set of weights that implement the trigonometric identity. Weight decay penalises large weights and therefore favours this compact solution over memorisation, which requires large weights to fit individual examples.

In our three-operand setting the identity may generalize to a direct closed-form expansion in the Fourier components of all three operands simultaneously:

$$\cos\left(\frac{2\pi k(a + b + c)}{p}\right) = \cos_a \cos_b \cos_c - \sin_a \sin_b \cos_c - \sin_a \cos_b \sin_c - \cos_a \sin_b \sin_c$$

$$\sin\left(\frac{2\pi k(a+b+c)}{p}\right) = \sin_a \cos_b \cos_c + \cos_a \sin_b \cos_c + \cos_a \cos_b \sin_c - \sin_a \sin_b \sin_c$$

where $\cos_a = \cos(2\pi ka/p)$ and so on. This raises the question of whether the transformer actually computes this direct formula, or whether it instead decomposes the problem into sequential partial sums $a+b$ and $(a+b)+c$, which would produce a qualitatively different internal circuit. We investigate this directly in the following section.

3 Implementation

3.1 Model and Training

We train a 2-layer transformer on $(a+b+c) \bmod 37$ from tokenised inputs of the form `a b c =`. The vocabulary has 38 tokens (residues $0\ldots36$ plus `=`). We use $d_{\text{model}} = 128$, 4 attention heads, feedforward dimension 512, no weight tying between input embeddings and the unembedding matrix, and compare both a causal self-attention and a non causal attention.

Training uses AdamW with learning rate 10^{-3}, $\beta = (0.9, 0.98)$, and weight decay 1.0, annealed via cosine scheduling to 10^{-4} over at least 15,000 steps in each run. At each step we sample a minibatch of size n_{train} with replacement from the training set, giving SGD with the same effective size as full batch but gradient noise that acts as a regularizer. The final mechanistic analysis was done on a model trained using 20% of all $37^3 = 50{,}653$ inputs.

3.2 Mechanistic Analysis Tools

Checkpoints are saved every 1,000 steps, each storing the full model weights and a cached forward pass on 512 held-out examples, including residual stream activations at every layer and attention weight matrices. All mechanistic analyses read directly from these caches without re-running the model. We use four complementary tools, each designed to answer a specific question about the learned algorithm.

Fourier energy directly measures the structure of the token embedding matrix $W_E \in R^{37\times128}$ by projecting each column onto a normalised Fourier basis and computing the energy at each frequency $k = 1, \ldots, 18$. This asks purely whether the embedding vectors, viewed as signals over the 37 tokens, oscillate at specific frequencies. Spikes at specific frequencies mean the model has chosen to represent numbers as points on circles, which is the prerequisite for implementing modular arithmetic via trigonometric identities.

Linear probes ask whether a specific quantity, such as the answer $a+b+c$, or a partial sum $a+b$, is linearly decodable in each position from the residual stream after a given transformer block. Probing for $\cos(2\pi kt/p)$ and $\sin(2\pi kt/p)$ directly tests whether the residual stream contains the fourier components the model is supposed to use in its computation. For each frequency k we fit an OLS regression from the 128-dimensional residual stream to $\cos(2\pi kt/p)$ and $\sin(2\pi kt/p)$ separately , reporting $R^2_{\cos} + R^2_{\sin} \in [0, 2]$. Values near 2 mean the full Fourier representation of the target is present; values near 0 mean it is absent. By probing different combinations of block, position, target, and frequency we try to map where and when each piece of information appears in the computation, and distinguish between individual operand encodings, partial sums, and the full answer.

Logit lens applies the model's own output LayerNorm + W_U decoder to intermediate residual streams and measures accuracy against the correct answer. This is different than the Linear probe: where the probe fits a new linear function and tests whether the information is decodable in principle, the logit lens uses the model's actual output head and tests whether the representation is already in the specific format W_U expects. A high Fourier $\mathrm{R}^{\mathbf{2}}$ with low logit lens accuracy means the answer has been computed but is not yet in output format (as it does not encode the result in all frequencies needed to reconstruct the output).

Activation patching is the only causal tool in our analysis. The three tools above are purely correlational, they tell us what information is present but not whether it matters for the output. To test causation we corrupt a single input operand $a \to a'$ and run both the clean input (a, b, c) and the corrupt input (a', b, c) through layer 0. We then replace the residual stream at exactly one token position with the corrupt version and feed this modified residual into layer 1, measuring whether the output shifts from $(a+b+c) \bmod p$ to $(a'+b+c) \bmod p$. If a position carries a's contribution causally, the output flips; if the position is inert, the output is unchanged. By repeating this for all four positions independently we aim to identify the causal pathway through which each operand contributes to the final answer, and directly test the staged computation hypothesis.

4 Results

4.1 Effect of Training Fraction and Weight Decay

Figure 5 shows validation and training accuracy across a grid of training fractions and weight decay values, with 3 random seeds per cell. The results confirm and extend the findings of Power et al. [2].

Weight decay's effect: With weight decay 0.01 the model memorises the training set rapidly (training accuracy reaches 1.0 within a few epochs) but validation accuracy remains near zero across almost all training fractions, with the exception of the largest training fractions. This confirms that without sufficient L2 regularisation the model settles for a high-norm memorisation solution and is never incentivized to discover the compact Fourier algorithm. Weight decay creates the pressure needed to abandon memorisation in favour of a lower-norm generalising solution.

Training fraction's effect: With weight decay 1 or 3, generalisation occurs reliably across all training fractions, but the delay before generalisation grows as the training fraction decreases. At frac = 0.6 validation accuracy jumps almost immediately; at frac = 0.05 the model may require the full 30,000 epochs or stronger weight decay to generalise. This is consistent with the intuition that a smaller training set is easier to memorise, so the model spends longer in the memorisation phase before weight decay compresses it toward the generalising solution.

High training fractions show direct generalisation: At frac ≥ 0.3 with sufficient weight decay, training and validation accuracy rise together without a prolonged memorisation plateau. This suggests that when the model sees enough data it does not find memorisation attractive in the first place and converges directly to an algorithmic solution, since memorizing a large amount of data would probably be as challenging as finding the algorithm. The grokking delay is therefore not an intrinsic property of the learning algorithm but a consequence of the tension between memorisation and generalisation that only arises when the training set is small enough to memorise cheaply.

Based on this grid search we selected frac = 0.2 and weight decay 1.0 for all subsequent mechanistic analyses. This configuration grokked reliably across all three seeds and exhibited a clean memorisation plateau followed by a sharp generalisation transition in both the causal and non-causal variants. Figure 1 shows the training dynamics for both selected models. The non-causal model grokks slightly earlier than the causal one, consistent with bidirectional attention providing more information per layer and reducing the delay before the Fourier algorithm is discovered. In both cases the weight norm decreases continuously before the accuracy jump, confirming that grokking reflects a gradual compression of the solution under regularisation pressure rather than a truly sudden transition.

4.2 Mechanistic Analysis of $(a + b + c) \bmod 37$

The central question of our mechanistic analysis is: what algorithm does the transformer learn, and how is the three-operand sum computed internally? We investigate this through Fourier probes, logit lens analysis, and causal interventions on the residual stream. To understand whether the learned algorithm depends on the presence of the causal mask, we train two models under identical conditions differing only in whether a causal mask is applied during attention. The causal model enforces that each position can only attend to earlier positions; the non-causal model allows full bidirectional attention. Both models grok and reach perfect validation accuracy, but the circuits they learn differ in instructive ways.

4.2.1 Does the Model Encode Numbers as Points on a Circle?

The first question is whether either model learns structured internal representations or simply memorizes them. We examine the token embedding matrix W_E directly. Before grokking, the Fourier energy

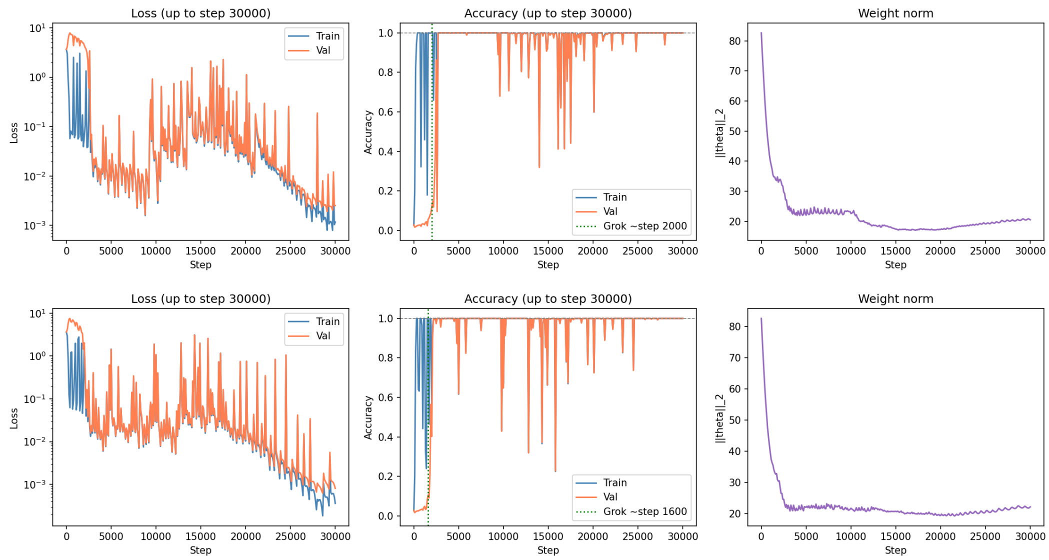

Figure 1: Training dynamics for the selected configuration (frac = 0.2, weight decay 1.0) for the causal model (top) and non-causal model (bottom). Each panel shows training loss (log scale), classification accuracy, and weight norm $\|\theta\|_2$ over 30,000 training steps.

of W_E is flat across all frequencies and the embeddings scatter without structure in PCA space, consistent with a memorizing model that does not need mathematical structure. At the grokking transition the picture changes: energy concentrates at a sparse set of frequencies and the tokens develop a clear geometric structure in the top two principal components (Figure 6).

The shape does not need to be a circle if multiple frequencies are active simultaneously. Each active frequency k defines its own circle in a separate two-dimensional subspace of the 128-dimensional embedding space, and PCA projects a mixture of these subspaces onto the plane.

The two models converge to different dominant frequencies in the reported run, the causal model mainly uses $k = 11$ and $k = 5$; while the non-causal model uses $k = 15$ and $k = 3$ (all in Figure 6) . This frequency selection is seed-dependent and not structurally meaningful; what matters is that both models independently discover the same class of solution. Under continued weight decay the active frequency set may narrow in the model, converging toward a sparser representation as the model finds the most sparse solution consistent with correct answers.

4.2.2 What is each layer's role?

The Fourier probe finds that the full sum $a + b + c$ is linearly decodable from the residual stream at position = after layer 0 shortly after grokking in both models, with $R^2 \approx 1.6$–1.8 at the dominant active frequencies and negative values at non-active ones. This is consistent with the frequency sweep shown in Figure 2, which displays the R^2 of the target a+b+c mod 37 across all frequencies $k = 1 \ldots 18$ at `resid_post_0` for each checkpoint: all bars are grey or negative before grokking, and a discrete set snaps into positive simultaneously at the grokking transition.

An important subtlety visible in both Figure 2 and Table 1 is that the decodability is tied to whichever frequencies happen to be active at that training step. At step 5,000 (just after grokking) the answer is decodable at $k = 1$ ($R^2 = 1.179$), but by step 15,000 the $k = 1$ signal has become strongly negative (-1.965) while $k = 11$ and $k = 15$ become dominant. The internal basis in which the result is represented continues to evolve after generalisation is achieved, even though accuracy remains at 1.000 throughout (apart from momentary instabilities).

The logit lens (Figure 3) shows that despite this signal being present, the output head cannot decode the answer from the layer 0 residual stream in either model (accuracy = 0.125 for the causal model, similar for the non-causal model, both barely above random chance $1/37 \approx 0.027$). Only after layer 1 does accuracy reach 1.000 in both cases. This dissociation, high Fourier R^2 but near-zero logit lens accuracy, might mean layer 0 computes the answer in a Fourier basis that layer 1 then rearranges

Step	$k=11$	$k=15$	$k=1$	$k=13$
0	-1.118	-1.249	-1.164	-0.968
1,000	-1.512	-1.300	-1.087	-1.313
2,000	-1.390	-1.011	-1.081	-1.333
5,000	-0.974	-0.297	$+1.179$	-0.455
10,000	$+0.553$	$+1.643$	-1.386	-0.772
15,000	$+1.962$	$+1.638$	-1.965	-1.473

Table 1: Fourier probe $R^2_{\text{cos}} + R^2_{\text{sin}}$ at position `=` after layer 0, target $a+b+c$, causal model.

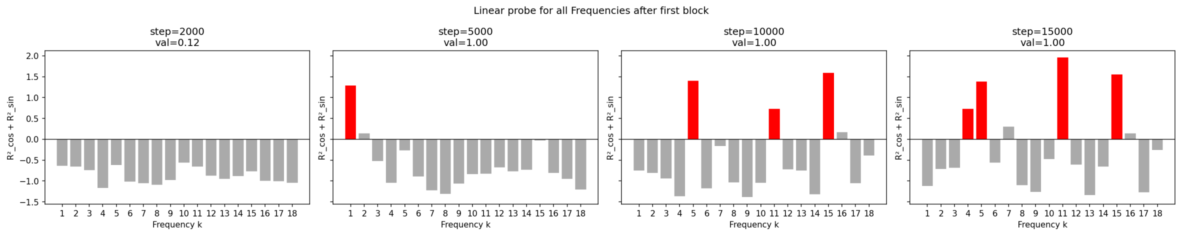

Figure 2: Fourier linear probe $R^2_{\text{cos}} + R^2_{\text{sin}}$ at `resid_post_0`, position `=`, target $a+b+c$, across training checkpoints for the causal model. Each bar corresponds to one frequency $k = 1 \dots 18$; red bars indicate $R^2 > 0.5$ (active), grey bars indicate inactive or suppressed frequencies.

into the format W_U expects. We did not investigate this further, but we speculate that the final block might need more frequencies to represent a clear spike to feed into the softmax. The non-causal model exhibits the same qualitative behaviour and is shown in the same Figure.

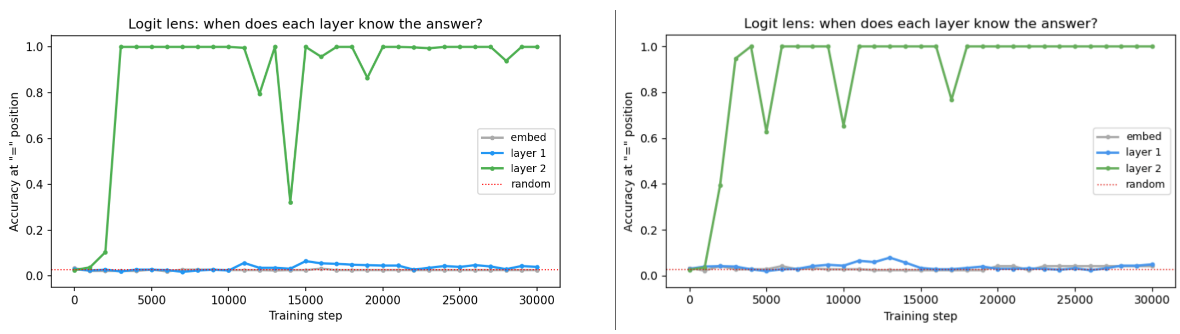

Figure 3: Logit lens accuracy at position `=` for target $a+b+c$ over training, for the causal model (left) and non-causal model (right). Each line shows the accuracy obtained by applying the model's own LayerNorm + W_U decoder to the residual stream at a given intermediate layer: after the embedding (grey), after layer 0 (blue), and after layer 1 (green). The red dotted line marks random chance ($1/37 \approx 0.027$). In both models the Fourier signal is present after layer 0 but not output-readable; only layer 1 completes the format conversion.

The attention patterns of both layers are consistent with this trend in both models: after grokking all four heads show roughly uniform attention from `=` to `a`, `b`, and `c` simultaneously. Figure 4 shows one representative head at three training stages: before grokking, at grokking, and late training, as all four heads in both layers show very similar patterns.

However there is a notable difference: in the causal model, the Fourier probe detects $R^2 \approx 1.1$ for target $a+b$ at position `b` and `c` after layer 0. Instead, in the non causal model, we see that all partial sums appear linearly decodable from the complementary missing operand. The reason could be structural: in the causal model, `b` is the first position that can see both `a` and itself, so it accumulates a mixture of their Fourier components as a byproduct of attention. In the non-causal model information routing is symmetric and by late training we see that every position encodes the full answer a+b+c.

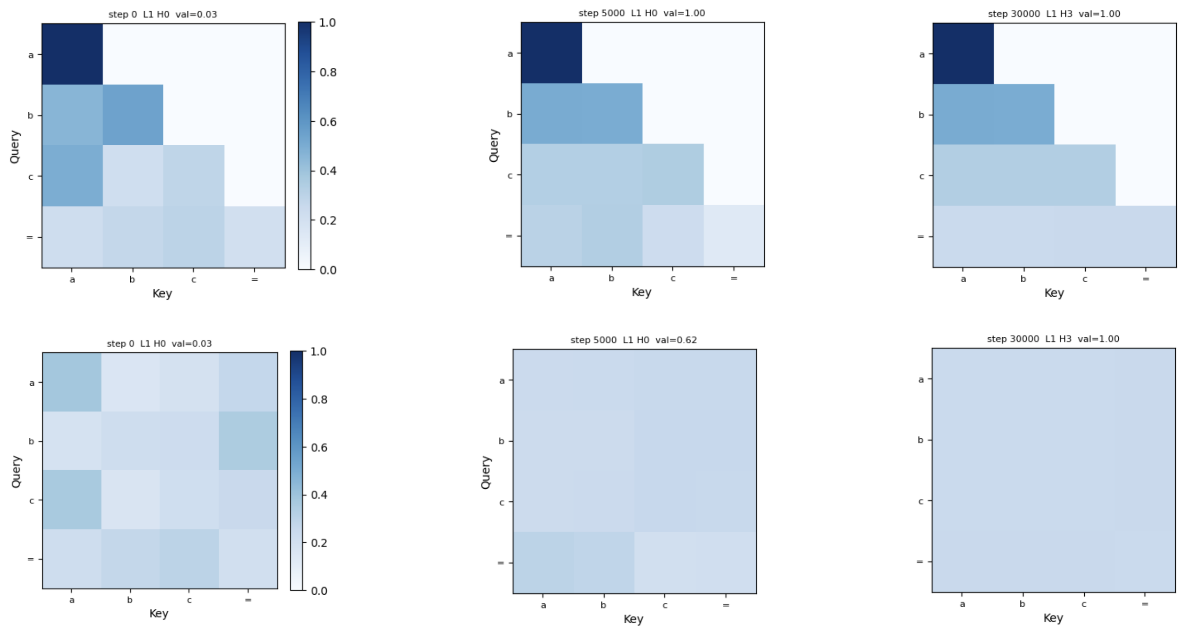

Figure 4: Layer 0 attention weights for one representative head at three training stages: before grokking (left), at grokking (centre), and late training (right). Top row: causal model. Bottom row: non-causal model.

4.2.3 Is partial computation performed?

The most revealing difference between the two models emerges from activation patching (Table 2). We corrupt a single operand $a \to a'$ and replace the residual stream at each position independently after layer 0, measuring whether the output shifts to $(a' + b + c) \bmod p$.

Both models grok at approximately step 3,000, after which both achieve perfect validation accuracy. However their patching results diverge significantly in the post-grokking consolidation phase.

In the **causal model** at step 15,000 the circuit has fully consolidated: replacing position `=` completely flips the output (acc_corrupt $= 1.000$), while replacing `a`, `b`, or `c` destroys the clean answer but never produces the corrupt one (acc_corrupt $= 0.000$ for all three). This indicates that all information is flowing through `=`.

In the **non-causal model** no consolidation at `=` occurs. By step 15,000, patching `=` has essentially no causal effect (acc_corrupt $= 0.168$, near random chance), while patching any of the three operand positions `a`, `b`, or `c` substantially damages the clean answer (acc_clean ≈ 0.65–0.76). The computation has distributed symmetrically across all operand positions rather than concentrating at `=`. Without the causal mask the model has no architectural incentive to funnel information through any single position, and the solution it finds distributes the computation symmetrically across the sequence.

5 Conclusions

We studied grokking in 2-layer decoder-only transformers trained on $(a+b+c) \bmod 37$, inspired by the mechanistic interpretability analysis of Nanda et al. [1], and comparing causal vs non-causal attention variants.

Our training dynamics experiments confirm the findings of Power et al. [2] in a three-operand setting: weight decay is the critical driver of grokking, and smaller training fractions delay but do not prevent generalization.

Our mechanistic analysis shows that both models independently discover similar Fourier circuits: token embeddings organize into a sparse set of frequencies, layer 0 assembles the full Fourier representation of $a+b+c$, at position `=` in the causal model, and distributed across all operand positions in the non-causal model. Activation patching rules out staged computation through intermediate positions in both models. Crucially, the spurious $R^2 \approx 1.2$ signal for $a+b$ at position `b` in the causal model,

	Causal			Non-causal		
Patched pos.	Effect	acc($a' + b + c$)	acc($a + b + c$)	Effect	acc($a' + b + c$)	acc($a + b + c$)
Step 5,000 — just after grokking, val_acc = 1.000 *for both*						
`a`	−3.64	0.098	0.436	−0.951	0.108	0.158
`b`	−10.25	0.000	1.000	−8.251	0.002	0.660
`c`	−10.26	0.000	1.000	−8.230	0.004	0.632
`=`	+3.57	0.488	0.096	+0.317	0.132	0.112
Step 15,000 — post-consolidation, val_acc = 1.000 *for both*						
`a`	−9.87	0.000	1.000	−9.417	0.030	0.658
`b`	−9.98	0.000	1.000	−10.303	0.014	0.762
`c`	−9.97	0.000	1.000	−10.149	0.016	0.748
`=`	+9.90	1.000	0.000	−0.245	0.168	0.142

Table 2: Activation patching results at step 5,000 (just after grokking) and step 15,000 (post-consolidation) for both models.

which could suggest staged computation, is absent in the non-causal model, providing independent evidence that it is an artefact of the causal mask rather than genuine intermediate computation.

While our analysis identifies the broad structure of the circuit, many questions remain open. A complete mechanistic account would require identifying whether different attention head components are needed for the representation of the trigonometric products or whether they can be collapsed into a single one, what the Fourier spectrum of the unembedding matrix W_U is, and whether the same structures emerge across different seeds, moduli, and numbers of operands. The post-grokking circuit drift under weight decay and the qualitative difference in circuit localisation between causal and non-causal models both deserve further investigation. We leave these directions for future work.

References

[1] Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. *arXiv preprint arXiv:2301.05217*, 2023.

[2] Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022.

Appendix

A. Training Dynamics Grid Search

Figure 5 shows the full grid search over training fraction and weight decay values used to select the experimental configuration for the mechanistic analysis. Each cell contains three runs with different random seeds, shown as separate curves.

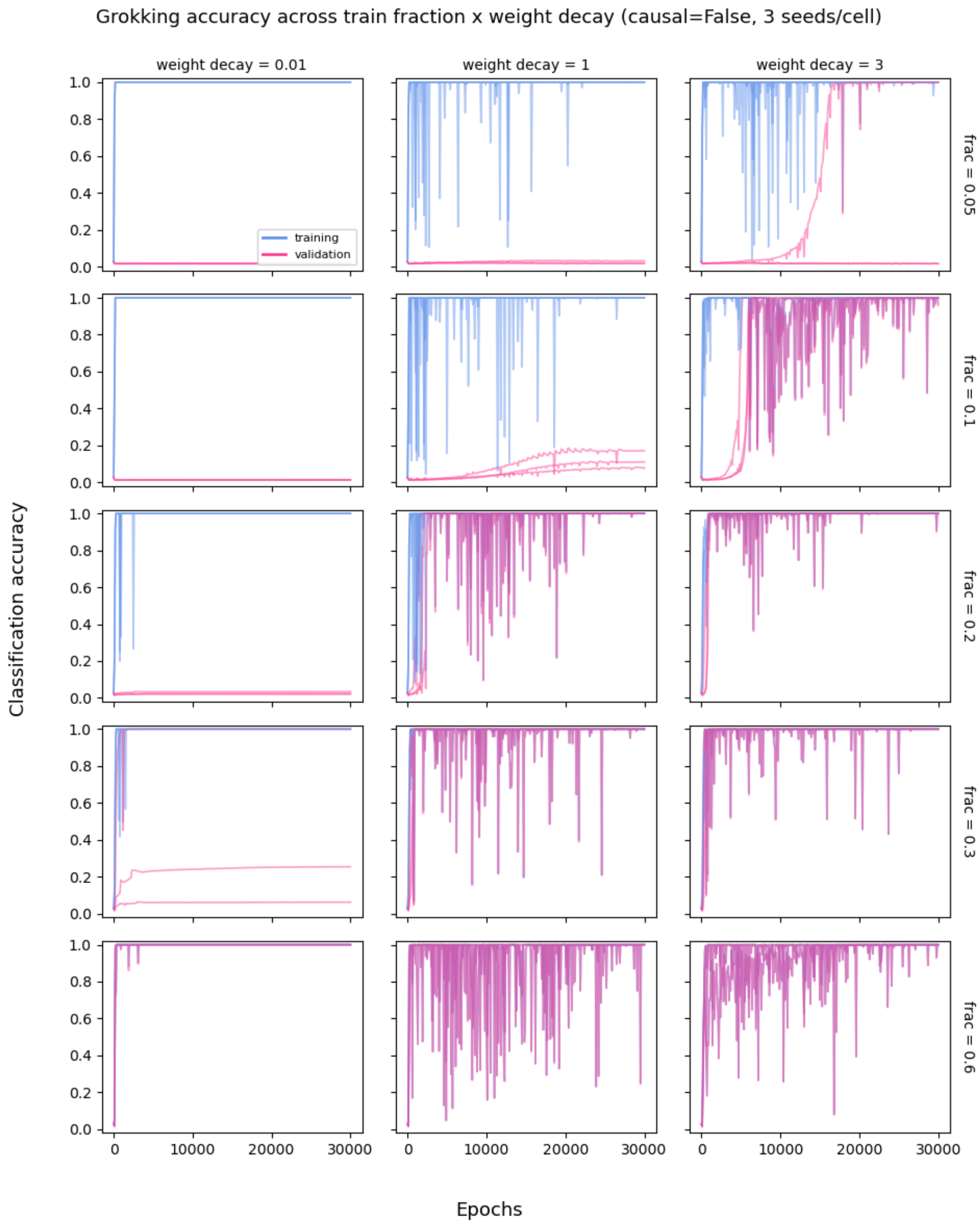

Figure 5: Validation and training accuracy across a grid of training fractions (0.05, 0.1, 0.2, 0.3, 0.6, rows) and weight decay values (0.01, 1.0, 3.0, columns), with 3 random seeds per cell (non-causal model, 30,000 epochs). Blue curves show training accuracy, pink curves show validation accuracy.

B. Embedding Structure: PCA and Fourier Energy

Figure 6 shows the token embedding PCA and Fourier energy spectra at step 30,000 for both models, after the circuit has fully consolidated under weight decay.

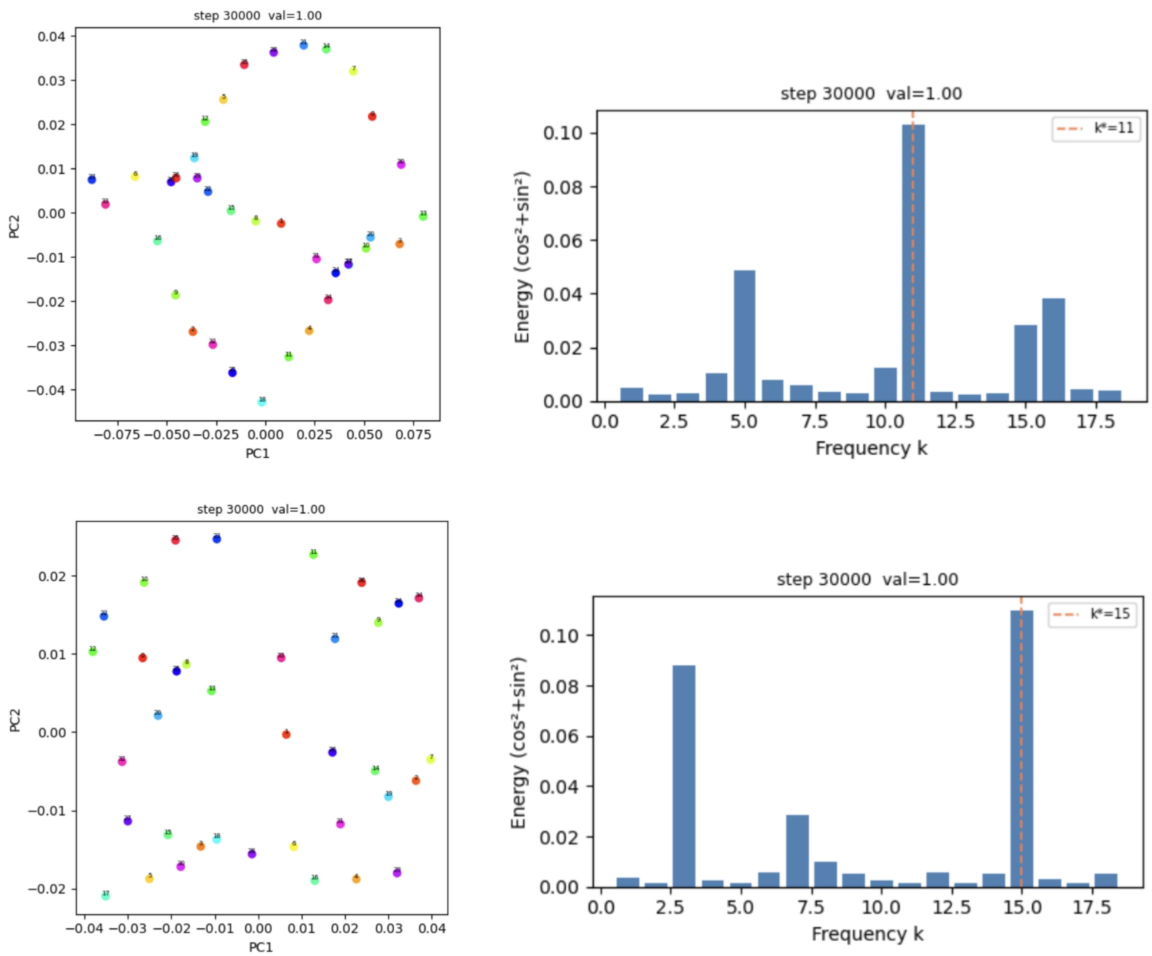

Figure 6: Token embedding PCA (left) and Fourier energy (right) at step 30,000 for the causal model (top) and non-causal model (bottom). Both models show structured geometric embeddings and sparse Fourier peaks after consolidation, converging to different dominant frequencies.